STAT 432 Homework 08

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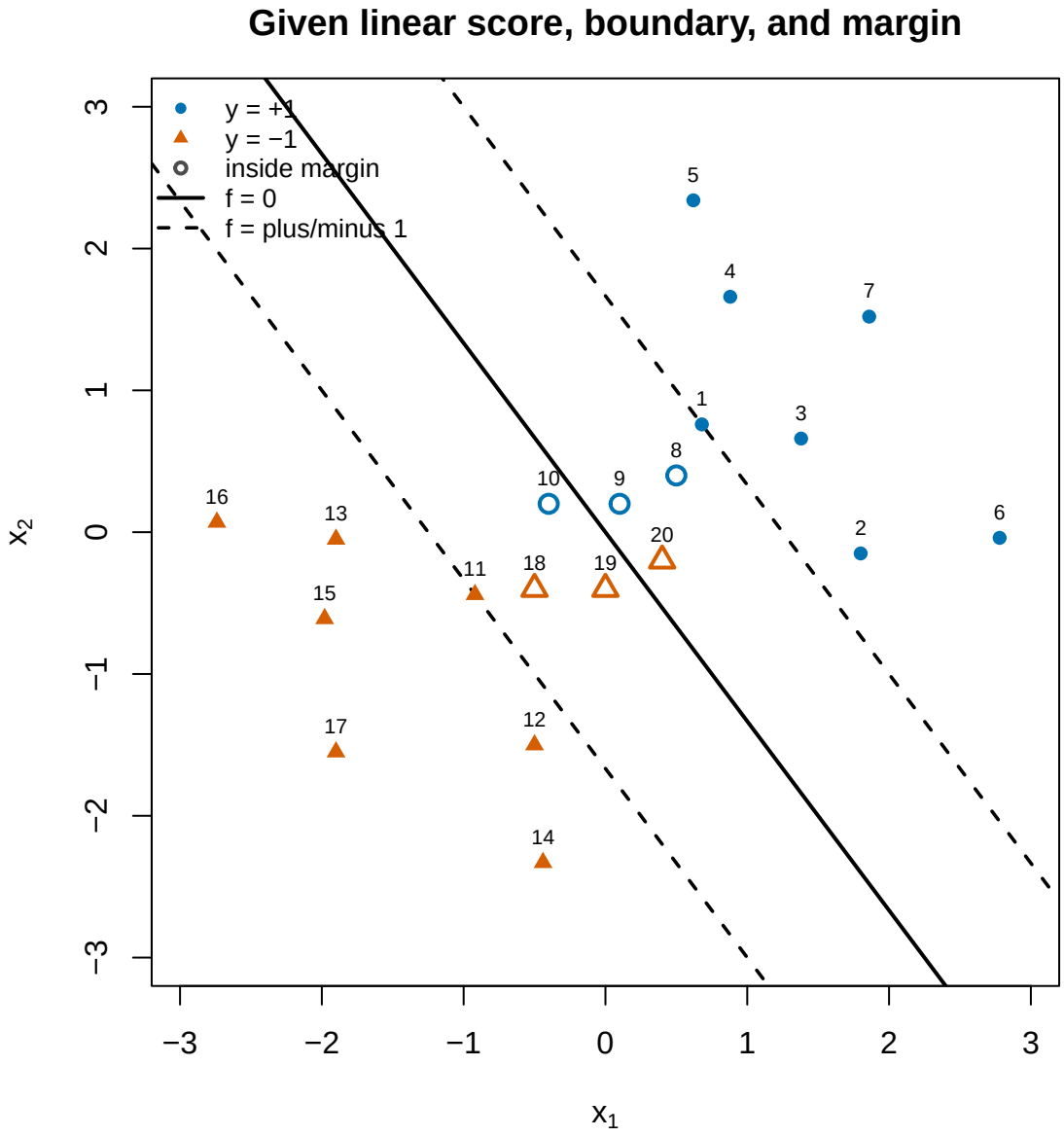
Question 1: Boundary and margin

The score is $f(x) = 0.8x_1 + 0.6x_2$ with $\|(0.8, 0.6)\|_2 = 1$, so the geometric distance to $f(x) = 0$ is $|f(x)|$. The functional margin is $yf(x)$. An observation is beyond the margin if $yf(x) > 1$, on the margin if $yf(x) = 1$, and inside if $yf(x) < 1$. It is misclassified if $yf(x) < 0$.

```
d <- read.csv("data/margin-geometry-20.csv")
d$f <- 0.8 * d$x1 + 0.6 * d$x2
d$dist <- abs(d$f)
d$yf <- d$y * d$f
d$where <- ifelse(d$yf > 1, "beyond",
                  ifelse(abs(d$yf - 1) < 1e-12, "on", "inside"))
d$misclassified <- d$yf < 0
data.frame(
  obs = d$observation,
  x1 = d$x1,
  x2 = d$x2,
  y = d$y,
  distance = round(d$dist, 2),
  yf = round(d$yf, 2),
  margin = d$where,
  misclassified = d$misclassified
)
```

```
##    obs    x1    x2  y distance    yf margin misclassified
## 1    1  0.68  0.76  1     1.00  1.00     on         FALSE
## 2    2  1.80 -0.15  1     1.35  1.35 beyond         FALSE
## 3    3  1.38  0.66  1     1.50  1.50 beyond         FALSE
## 4    4  0.88  1.66  1     1.70  1.70 beyond         FALSE
## 5    5  0.62  2.34  1     1.90  1.90 beyond         FALSE
## 6    6  2.78 -0.04  1     2.20  2.20 beyond         FALSE
## 7    7  1.86  1.52  1     2.40  2.40 beyond         FALSE
## 8    8  0.50  0.40  1     0.64  0.64 inside         FALSE
## 9    9  0.10  0.20  1     0.20  0.20 inside         FALSE
## 10  10 -0.40  0.20  1     0.20 -0.20 inside          TRUE
## 11  11 -0.92 -0.44 -1     1.00  1.00     on         FALSE
## 12  12 -0.50 -1.50 -1     1.30  1.30 beyond         FALSE
## 13  13 -1.90 -0.05 -1     1.55  1.55 beyond         FALSE
## 14  14 -0.44 -2.33 -1     1.75  1.75 beyond         FALSE
## 15  15 -1.98 -0.61 -1     1.95  1.95 beyond         FALSE
## 16  16 -2.74  0.07 -1     2.15  2.15 beyond         FALSE
## 17  17 -1.90 -1.55 -1     2.45  2.45 beyond         FALSE
## 18  18 -0.50 -0.40 -1     0.64  0.64 inside         FALSE
## 19  19  0.00 -0.40 -1     0.24  0.24 inside         FALSE
## 20  20  0.40 -0.20 -1     0.20 -0.20 inside          TRUE
```

```r
par(mar = c(4.2, 4.2, 3.0, 1.0), pty = "s")
plot(d$x1, d$x2, type = "n",
     xlim = c(-3.2, 3.2), ylim = c(-3, 3), asp = 1,
     xaxs = "i", yaxs = "i",
     xlab = expression(x[1]), ylab = expression(x[2]),
     main = "Given linear score, boundary, and margin")
abline(0, -0.8 / 0.6, lwd = 2)
abline(1 / 0.6, -0.8 / 0.6, lwd = 1.8, lty = 2)
abline(-1 / 0.6, -0.8 / 0.6, lwd = 1.8, lty = 2)
inside <- d$where == "inside"
pos <- d$y == 1
points(d$x1[pos & !inside], d$x2[pos & !inside], pch = 16, col = "#0072B2")
points(d$x1[!pos & !inside], d$x2[!pos & !inside], pch = 17, col = "#D55E00")
points(d$x1[pos & inside], d$x2[pos & inside], pch = 1, col = "#0072B2", cex = 1.3, lwd = 2)
points(d$x1[!pos & inside], d$x2[!pos & inside], pch = 2, col = "#D55E00", cex = 1.3, lwd = 2)
text(d$x1, d$x2, labels = d$observation, pos = 3, cex = 0.7)
legend("topleft", bty = "n", cex = 0.8,
       legend = c("y = +1", "y = -1", "inside margin",
                  "f = 0", "f = plus/minus 1"),
       pch = c(16, 17, 1, NA, NA),
       col = c("#0072B2", "#D55E00", "gray30", "black", "black"),
       lty = c(NA, NA, NA, 1, 2),
       lwd = c(NA, NA, 2, 2, 1.8))
```



(a)–(c)

The table is above. Counts: 12 beyond, 2 on (observations 1 and 11), and 6 inside (8, 9, 10, 18, 19, 20). Observations 10 and 20 have $yf(x) = -0.20$, so they are the only misclassified points. They sit on the wrong side of $f(x) = 0$ and therefore also count as inside the margin.

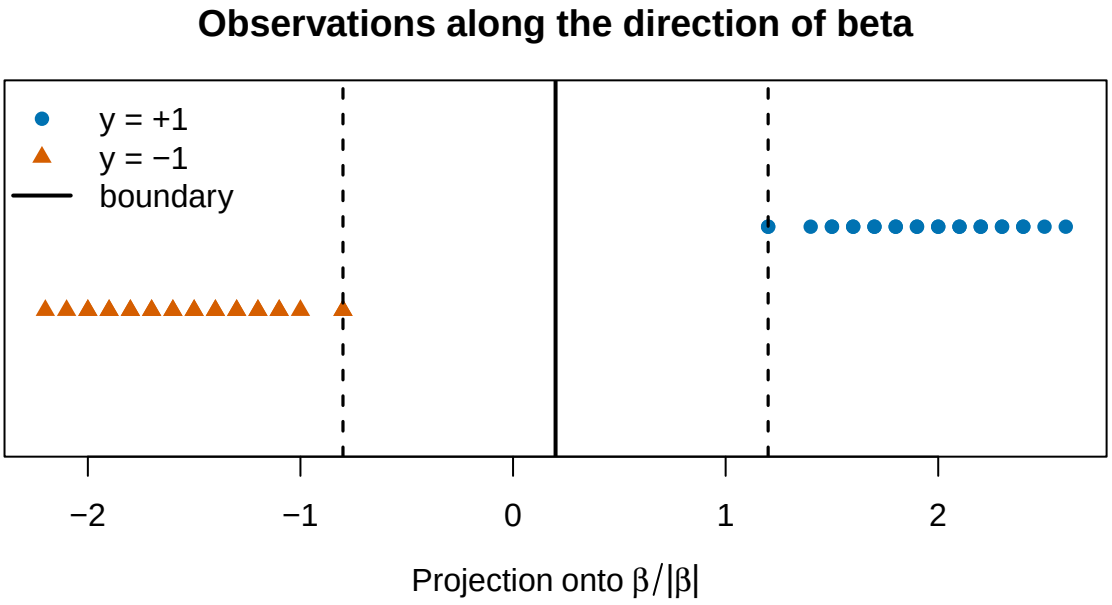
Question 2: Margin in four dimensions

```
d4 <- read.csv("data/margin-geometry-4d.csv")
b0 <- -1
b <- c(1, 2, -2, 4)
```

```r
X <- as.matrix(d4[, c("x1", "x2", "x3", "x4")])
f <- b0 + as.numeric(X %*% b)
nb <- sqrt(sum(b^2))
u <- b / nb
proj <- as.numeric(X %*% u)
boundary <- -b0 / nb
dist <- abs(f) / nb
M <- min(dist)
mfun <- min(d4$y * f)
c(norm_beta = nb, boundary = boundary, M = M, min_yf = mfun)
```

```
## norm_beta  boundary         M    min_yf
##       5.0       0.2       1.0       5.0
```

```r
par(mar = c(4.2, 2.8, 3.0, 1.0))
yg <- ifelse(d4$y == 1, 0.12, -0.12)
plot(proj, yg, pch = ifelse(d4$y == 1, 16, 17),
     col = ifelse(d4$y == 1, "#0072B2", "#D55E00"),
     ylim = c(-0.5, 0.5),
     xlab = expression(Projection ~ "onto" ~ beta / group("||", beta, "||")),
     ylab = "",
     yaxt = "n",
     main = "Observations along the direction of beta")
abline(v = boundary, lwd = 2)
abline(v = boundary + c(-M, M), lwd = 1.6, lty = 2)
legend("topleft", bty = "n",
       legend = c("y = +1", "y = -1", "boundary"),
       pch = c(16, 17, NA),
       col = c("#0072B2", "#D55E00", "black"),
       lty = c(NA, NA, 1), lwd = c(NA, NA, 2))
```



(a)

$\|\boldsymbol{\beta}\|_2 = 5$, so the boundary $f(\mathbf{x}) = 0$ sits at projection $-\beta_0/\|\boldsymbol{\beta}\|_2 = 0.2$. The closest points (observations 4, 15, 33, and 36) are distance 1 from that line, so the geometric margin is $M = 1$. All 50 points are correctly classified.

(b)

The smallest functional margin is $\min_i y_i f(\mathbf{x}_i) = 5$. Dividing the score by 5 keeps the same boundary and makes that minimum equal to 1:

$$\boldsymbol{\beta} = (0.2, 0.4, -0.4, 0.8)^{\mathsf{T}}, \qquad \beta_0 = -0.2.$$

Question 3: Peer review of a Homework 7 skill

I got assigned https://github.com/Gicooaidun/stat432-fall2026 (issue 3 in my repo). I copied their skill out of the Homework 7 PDF into `homework/week-08/peer-skill/SKILL.md` and did not install it. The only thing I asked the agent was:

> Follow `homework/week-08/peer-skill/SKILL.md` on `data/saheart.csv`. Response: `chd`. Predictors: the nine other columns, with `famhist` coded 1 for Absent and 2 for Present.

I did not add extra rules about plots or which model to pick. The report from that run is on the next page. I posted the same review at https://github.com/Gicooaidun/stat432-fall2026/issues/5.

Skill-directed SAheart report

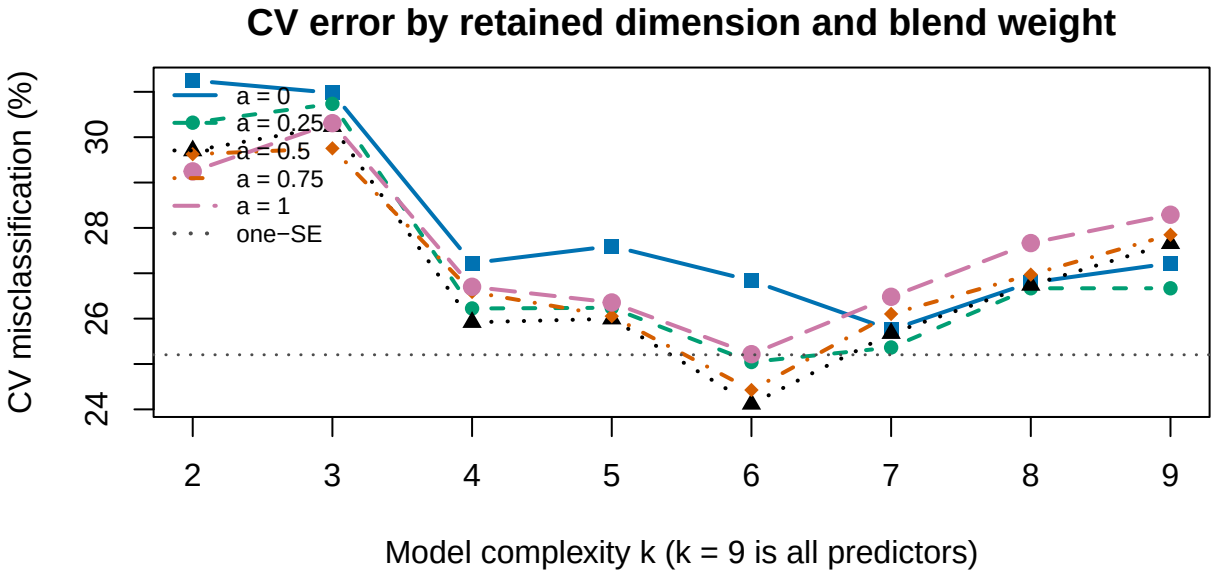


Table 1: CV error, fold SE, covariance-parameter count. Test error only for the selected model.

candidate	CV %	SE %	cov. params	test %
LDA, all 9	27.2	1.0	45	NA
QDA, all 9	28.3	1.2	90	NA
min CV: PCA k=6 blend a=0.5	24.1	1.1	42	NA
selected: PCA k=6 blend a=0.5	24.1	1.1	42	28.3

Setup. $n = 462$, $p = 9$, classes 302 and 160 (`chd` 0/1, `famhist` 1/2). Seed `43208`, stratified 30% test (324 train, 138 test). All candidates share five repeats of 10-fold stratified CV. Standardization and PCA are fit inside each training fold only. No fit was singular.

Numerical results. The minimum CV error is PCA $k = 6$ with blend $a = 0.5$: 24.1% (SE 1.1%). The one-SE set has three candidates, all $k = 6$ blends with 42 covariance parameters, so the 1-SE rule selects the same model. Test error is 28.3% (39/138). Confusion matrix: predicted 0/1 against observed 0: 75, 15; observed 1: 24, 24.

Findings. Ordinary QDA on all nine predictors (28.3% CV) is worse than ordinary LDA (27.2%). The data do not favor a full per-class 9×9 covariance at this n and p. A blend at reduced dimension ($k = 6$) has the best CV rate, which is a cheaper quadratic term after PCA. The CV gap over LDA is about 3 points, a little over two fold SEs, but that model was not simpler than LDA on the parameter count.

Limitations. The test SE is $\sqrt{0.283(1 - 0.283)/138} \approx 3.8$ points, so the selected model's test error is within noise of a 27% rate. LDA and QDA assume within-class normality; `tobacco` and `alcohol` are skewed. The 0.5 cutoff ignores the 35% positive rate.

Peer review

The skill is pretty clear. It only allows LDA/QDA (and PCA or a blend of those), it splits the data first, and it reuses the same folds. I copied it as-is and ran it on SAheart. It did produce a one-page report with a figure and a table, so the main protocol works. A few things I'd change:

1. Blends never look "simple." I counted covariance parameters the way the skill says, so a blend got the same count as QDA. In my run the 1-SE set was just three $k = 6$ blends, all with 42 parameters, and it picked the min-CV model. If 1-SE is supposed to prefer a cheaper fit, the skill should say whether a blend counts as LDA, QDA, or both added together.

2. A couple of defaults are missing. There is no seed, and it doesn't say how to turn the blended probabilities into a label. I used seed 43208 and picked the class with the larger probability. Someone else could follow the same skill and get a different model. I'd just put the seed in the prompt and say what cutoff to use.

3. The Homework 7 table doesn't quite match the skill. The skill says to touch the test set once, but their table has test error for LDA, QDA, and the selected model. That makes it look like the test set was used to compare everything. I'd only report test error for the winner, or say LDA and QDA are just reference numbers and were not used to pick the model.

Same text: https://github.com/Gicooaidun/stat432-fall2026/issues/5